

GENERAL SPECIFICATION G5.51

Electrical Testing

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G5.51 ELECTRICAL TESTING

G5.51.1 General

(a) References

- (i) The Contractor shall refer to General Specification G5.2 - Codes, Standards and Regulations.
- (ii) The following is a list of standards which may be referenced in this Section:
 - 1) British Standards Institute (BSI)

Standard	Subject
BS 89-9	Direct Acting Indicating Analogue Electrical Measuring Instruments and Their Accessories.
BS 159	High Voltage Bus bars and Bulbar Connections.
BS 764	Specification for Automatic Change-Over Contactors for Emergency Lighting Systems.
BS 799	Oil Burning Equipment.
BS 1212	Float Operated Valves.
BS 1449	Steel Plate, Sheet and Strip. Carbon and Carbon-Manganese Plate, Sheet and Strip.
BS 1322	Specification for Amino plastic Moulding Materials.
BS 1363	Specification for 13 A Plugs, Socket-Outlets and Connection Units.
BS 4678	Cable Trunking.
BS 5424	Specification for Control gear for Voltages Up to and Including 1,000 V a.c. and 1,200 V d.c. Contactors.
BS 5467	Specification for 600/1,000 V and 1,900/3,300 V Armoured Electric Cables Having Thermosetting Insulation.
BS 6004	Specification for PVC-Insulated Cables (Non-Armoured) for Electric Power and Lighting.
BS 6231	Specification for PVC-Insulated Cables for Switchgear and Control gear Wiring.
BS 6387	Specification for Performance Requirements for Cables Required to Maintain Circuit Integrity Under Fire Conditions.
BS 7430	Code of Practice for Protective Earthing for Electrical Installations.
BS EN 12285-2	Specification for Carbon Steel Welded Horizontal Cylindrical Storage Tanks.
BS EN60060-1	Guide on High Voltage Test Techniques.

Standard	Subject
BS EN 60076	Specification for Power Transformers.
BS EN 60079	Specification for Electrical Apparatus for Explosive Atmospheres. Classification of Maximum Surface Temperatures.
BS EN 60085	Electrical Insulation. Thermal Classification
BS EN 60228	Specification for Conductors in Insulated Cables and Cords.
BS EN 62271-100	High Voltage Alternating Current Circuit Breakers.
BS EN 62271-106	High Voltage Switchgear and Controlgear. Alternating Current Contactors, Contactor based Controller and Motor Starters.
BS EN 60309	Plugs, Socket-Outlets and Couplers for Industrial Purposes.
BS EN 60376	Specification for Sulphur Hexafluoride for Electrical Equipment.
BS EN 60480	Code of Practice for the Testing of Sulphur Hexafluoride Taken From Electrical Equipment.
BS EN 60669-1	Switches for Household and Similar Fixed Electrical Installations. Specification for General Requirements.
BS EN 60702-3	Mineral Insulated Cables with a Rated Voltage Not Exceeding 750 V.

2) International Electro-technical Commission (IEC)

Standard	Subject
IEC 60034	Rotating Electrical Machines.
IEC 60044	Instrument Transformers.
IEC 60050	International Electro-technical Vocabulary.
IEC 60051	Direct Acting Indicating Analogue Electrical Measuring Instruments and Their Accessories.
IEC 60060	High Voltage Test Techniques.
IEC 60076	Power Transformers.
IEC 60079	Electrical Apparatus for Explosive Gas Atmospheres.
IEC 60083	Plug and Socket-Outlet for Domestic and Similar General Use Standardised in Member Countries of IEC.
IEC 60085	Thermal Evaluation and Classification of Electrical Installation.

Standard	Subject
IEC 60086	Primary Batteries.
IEC 60137	Insulating Bushings for Alternating Voltages Above 1,000 V.
IEC 60168	Tests On Indoor and Outdoor Post Insulators of Ceramic Material or Glass for Systems with Nominal Voltages Greater Than 1,000 V.
IEC 60269	Low Voltage Fuses.
IEC 60273	Characteristic of Indoor and Outdoor Post Insulators for Systems With Nominal Voltages Greater Than 1,000 V.
IEC 60309	Plug, Socket-Outlet and Coupler for Industrial Purposes.
IEC 60364	Electrical Installations of Buildings.
IEC 60480	Guide to the Checking of Sulphur Hexafluoride (SF ₆) Taken From Electrical Equipment.
IEC 61442	Electric Cables Test Methods for Accessories for Power Cables with Rated Voltages From 6 kV (Um = 7.2 kV) Up to 30 kV (Um = 36 kV).
IEC 61643-1	Surge Protective Devices Connected to Low Voltage Power Distribution Systems – Part 1: Performance Requirements and Testing Methods.
IEC 60502-1	Power Cables With Extruded Insulation and Their Accessories for Rated Voltages From 1 kV (Um = 1.2 kV) Up to 30 kV (Um = 36 kV) – Part 1: Cables for Rated Voltages of 1 kV (Um = 1.2 kV) and 3 kV (Um = 3,6 kV).
IEC 60502-2	Power Cables With Extruded Insulation and Their Accessories for Rated Voltages From 1 kV (Um = 1.2 kV) Up to 30 kV (Um = 36 kV) – Part 2: Cables for Rated Voltages From 6 kV (Um = 7.2 kV) Up to 30 kV (Um = 36 kV).
IEC 60502-4	Power Cables With Extruded Insulation and Their Accessories for Rated Voltages From 1 kV (Um = 1.2 kV) Up to 30 kV (Um = 36 kV) – Part 4: Test Requirements on Accessories for Cables with Rated Voltages From 6 kV (Um = 7.2 kV) Up to 30 kV (Um = 36 kV).
IEC 60529	Degrees of Protection Provided by Enclosures (IP Code).
IEC 60617	Graphical Symbols for Diagrams.
IEC 60811	Common Test Methods for Insulating and Sheathing Materials of Electric Cables.

Standard	Subject
IEC 60840	Power Cables with Extruded Insulation and Their Accessories for Rated Voltages Above 30 kV (Um = 36 kV) Up to 150 kV (Um = 170 kV) - Test Methods and Requirements.
IEC 60885	Electrical Test Methods for Electric Cables.
IEC 60896	Stationary Lead-Acid Batteries General Requirements and Test Methods.
IEC 60898	Electrical Accessories - Circuit-Breakers for Overcurrent Protection for Household and Similar Installations.
IEC 60947	Low Voltage Switchgear and Control gear.

3) Spring Singapore (SS)

Standard	Subject
CP 5	Code of Practice for Electrical Installation.
CP 83	Construction Computer-Aided Design.
SS 97	Residual Current Circuit Breakers (RCCBs)
SS 145	13 A Plugs, Switched and Unswitched Socket Outlets.
SS 299	Fire Resistant Cables.
SS 322	Earthing and Bonding Clamps.
SS 516	Application, Operation and Maintenance of Dry-Type Transformers.
SS 551	Code of Practice for Earthing.
SS 538	Code of Practice for Maintenance of Electrical Equipment of Electrical Installations.
SS 535	The Installation, Operation, Maintenance, Performance and Constructional Requirements of Mains Failure Standby Generating Systems.
SS 555	Lightning Protection.

(iii) The following is a list of local statutory requirements to comply with:

- 1) EMA regulations;
- 2) Electricity Act;
- 3) Factories Act (Chapter 104), Singapore;
- 4) Fire Safety Act, Singapore; and
- 5) All other local statutory acts that are in place at the time of construction.

(b) Submittals

- (i) Submit the following **30** days prior to performing inspections or tests (All electrical tests shall be carried out by Contractor's LEW with appropriate Engineer Grade and witnessed by S.O.):
 - 1) Schedule for performing inspection and tests;
 - 2) List of references to be used for each test;
 - 3) List of certified test equipment used with details of Brand and model;
 - 4) Sample copy of equipment and materials inspection form(s);
 - 5) Sample copy of individual device test form; and
 - 6) Sample copy of individual system test form.
- (ii) Submit the following within 30 days after completion of test:
 - 1) Test or inspection reports and certificates for each electrical item tested, and
 - 2) Operation and Maintenance Manual:
 - After test or inspection reports and certificates have been reviewed by S.O. and returned, insert a copy of each in Operation and Maintenance Manual.

(c) Quality Assurance

- (i) Testing Firm Qualifications
 - 1) Corporately and financially independent organisation functioning as an unbiased testing authority.
 - 2) Professionally independent of manufacturers, suppliers, and installers, of electrical equipment and systems being tested.
 - 3) Testing engineers and technicians regularly engaged in testing and inspecting of electrical equipment, installations, and systems.
 - 4) All cabling termination and jointing works shall be carried out by licensed cable jointers.
 - 5) Supervising engineer shall be Licensed Electrical Worker (LEW) registered in Singapore as a testing engineer and shall have a minimum of 5 years' testing experience on similar projects. Testing and Commissioning of all electrical equipment shall be carried out in presence of competent LEW.
 - 6) Registered professional engineer registered in Singapore to provide comprehensive project report outlining services performed, results of such services, recommendations, actions taken, and opinions.
- (ii) Test equipment shall have an operating accuracy equal to, or greater than, requirements established by SS or IEC or BS.
- (iii) Test instrument calibration shall be in accordance with requirements of applicable Singapore Standards, IEC Standards, and BSI Standards.

- (d) Sequencing and Scheduling
 - (i) Perform inspection and electrical tests after equipment has been installed.
 - (ii) Perform tests with apparatus de-energised whenever feasible. If any test to be carried out on the energized equipment, the Contractor's LEW shall ensure that electrical safety is not compromised.
 - (iii) Inspection and electrical tests on energised equipment are to be:
 - 1) Scheduled with S.O. prior to de-energisation; and
 - 2) Minimised to avoid extended period of interruption to the operating plant equipment.
 - (iv) Notify S.O. at least 48 hours prior to performing tests on energised electrical equipment.

G5.51.2 Execution

- (a) General
 - (i) Tests specified in this Section are to be performed in accordance with the requirements of General Specification G2.26 – Plant Testing and Commissioning.
 - (ii) Tests and inspection shall establish that:
 - 1) Electrical equipment is operational within industry and manufacturer's recommended tolerances for actual site applications. To ensure that the equipment operates in accordance with manufacturer's recommendations on safety.
 - 2) Installation operates properly.
 - 3) Testing and inspection shall be carried out by Contractor's LEW with appropriate grade. Contractor and their LEW shall ensure that the competent and trained technicians are engaged for testing and inspection.
 - 4) Equipment is suitable for energisation.
 - 5) Installation conforms to requirements of Contract Documents and requirements of applicable Singapore Standards, or IEC Standards.
 - (iii) Perform inspection and testing in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards or industry standards, and manufacturer's recommendations.
 - (iv) Set, test, and calibrate protective relays, circuit breakers and other applicable devices in accordance with values established by the short-circuit and co-ordination study as specified in General Specification G5.50 – Electrical Systems Analysis.
 - (v) Adjust mechanisms and moving parts for free mechanical movement.
 - (vi) Adjust adjustable relays and sensors to correspond to operating conditions, or as recommended by manufacturer.
 - (vii) Verify nameplate data for conformance to Contract Documents.
 - (viii) Realign equipment not properly aligned and correct un-levelness.

- (ix) Properly anchor electrical equipment found to be inadequately anchored.
- (x) Tighten accessible bolted connections, including wiring connections, with calibrated torque wrench to manufacturer's recommendations, or as otherwise specified.
- (xi) Clean contaminated surfaces with cleaning solvents as recommended by manufacturer.
- (xii) Provide proper lubrication of applicable moving parts.
- (xiii) Provide safety guards or proper cover for any moving parts that are dangerous and unsafe.
- (xiv) Investigate and repair or replace:
 - 1) Electrical items that fail tests;
 - 2) Active components not operating in accordance with manufacturer's instructions; and
 - 3) Damaged electrical equipment.
- (xv) Electrical Enclosures
 - 1) Remove foreign material and moisture from enclosure interior;
 - 2) Vacuum and wipe clean enclosure interior;
 - 3) Remove corrosion found on metal surfaces;
 - 4) Repair or replace, as determined by S.O., door and panel sections having dented surfaces;
 - 5) Repair or replace, as determined by S.O., poor fitting doors and panel sections;
 - 6) Repair or replace improperly operating latching, locking, or interlocking devices;
 - 7) Replace missing or damaged hardware;
 - 8) Replace missing external or internal labels, damaged seals, damaged cables, damaged intake and extraction fans, covered glad openings, or panel lightings with appropriate electrical accessories;
 - 9) Finish:
 - Provide matching paint and touch up scratches and mars; and
 - If required due to extensive damage, as determined by S.O., refinish the entire assembly.
- (xvi) Replace fuses and circuit breakers that do not conform to size and type required by the Specifications and/or Drawings.
- (b) Switchgear and Switchboard Assemblies
 - (i) Visual and Mechanical Inspection
 - 1) Insulator damage and contaminated surfaces;
 - 2) Proper barrier and shutter installation and operation;

- 3) Proper operation of indicating devices;
- 4) Improper blockage of air cooling passages;
- 5) Proper operation of draw out elements;
- 6) Integrity and contamination of bus insulation system;
- 7) Check Door and Device Interlocking System by:
 - Closure attempt of device when door is in **OFF or OPEN** position; and
 - Opening attempt of door when device is in **ON or CLOSED** position.
- 8) Check Key Interlocking Systems for:
 - Key captivity when device is in ON or CLOSED position;
 - Key removal when device is in ON or CLOSED position;
 - Closure attempt of device when key has been removed;
 - Correct number of keys in relationship to number of lock cylinders; and
 - Existence of other keys capable of operating lock cylinders: Destroy duplicate sets of keys.
- 9) Check Nameplates for Proper Identification of:
 - Equipment title and tag number with latest one-line diagram;
 - E-Stop Push Buttons;
 - Local/Remote selector switch and indicating lights;
 - Pilot light;
 - Control relay;
 - Circuit breaker;
 - Indicating metre; and
 - Power meters and relays.
- 10) Verify that fuse and circuit breaker ratings, sizes, and types conform to those specified.
- 11) Check bus and cable connections for high resistance by using low resistance ohmmeter and thermographic survey applied to bolted joints.
 - Ohmic value to be zero; and
 - Thermographic survey temperature gradient of 2 degrees C, or less.
- 12) Check Operation and Sequencing of Electrical and Mechanical Interlock Systems by Contractor's LEW by:
 - Closure attempt for locked open devices;
 - Opening attempt for locked closed devices;
 - Key exchange to operate devices in OFF-NORMAL positions;
 - Contractor shall provide step by step operation procedure and instructions; and

- The operational procedure shall be checked and endorsed by LEW.
- 13) Verify performance of each control device and feature.
- 14) Control Wiring:
- Compare wiring to local and remote control and protective devices with elementary diagrams;
 - Proper conductor lacing and bundling;
 - Proper conductor identification;
 - Proper conductor lugs and connections;
 - Contractor shall ensure that all wirings and components are properly labelled and terminated;
 - The Contractor shall provide detailed termination drawings to S.O. for checking and verification;
 - All unused cables are to be labelled as 'Not Used' and bundle together with proper cable ties; and
 - Exposed unused cables are not allowed.
- 15) Exercise active components.
- 16) Perform phasing check on double-ended equipment to ensure proper bus phasing from each source.
- 17) Additional test for SF₆ insulated switchgear:
- Test for gas leaks;
 - Test SF₆ gas for moisture and SO₂ content; and
 - Test SF₆ gas dewpoint or any other tests recommended by the manufacturers.
- (ii) Electrical Tests (Shall be carried out by LEW with appropriate grade and witnessed by S.O.'s representative):
- 1) Insulation Resistance Tests:
- Applied megohmmeter DC voltage in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards;
 - Each phase of each bus section;
 - Phase-to-phase and phase-to-earth for 1 minute;
 - With breakers open;
 - With breakers closed;
 - Control wiring except that connected to solid state components; and
 - Insulation resistance values equal to, or greater than, ohmic values established by manufacturer.

- 2) Over potential Tests:
 - Applied AC or DC voltage and test procedure in accordance with applicable IEC Standards;
 - Each phase of each bus section;
 - Phase-to-phase and phase-to-earth for 1 minute; and
 - Test results evaluated on a pass/fail basis.
 - 3) Current Injection Tests:
 - For entire current circuit in each section;
 - Secondary injection for current flow as specified for protective relays; and
 - Test current at each device.
 - 4) Control Wiring:
 - Apply secondary voltage to control power and potential circuits;
 - Check voltage levels at each point on terminal boards and each device terminal;
 - Contractor shall ensure that all wirings and components are properly labelled and terminated;
 - All unused cables are to be labelled as 'Not Used' and bundle together with proper cable ties;
 - Exposed unused cables are not allowed; and
 - Contractor shall provide termination drawing for checking and verification.
 - 5) Operational Test:
 - Initiate control devices; and
 - Check proper operation of control system in each section.
- (c) Liquid Filled Transformers
- (i) Visual and Mechanical Inspection (Shall be carried out by Contractor's LEW and witnessed by S.O.):
 - 1) Physical and insulator damage;
 - 2) Proper winding connections;
 - 3) Bolt torque level in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards unless otherwise specified by manufacturer;
 - 4) Defective wiring;
 - 5) Proper operation of fans, indicators, and auxiliary devices;
 - 6) Effective core and equipment earthing;
 - 7) Removal of shipping brackets, fixtures, or bracing;
 - 8) Tank leaks and proper liquid level;

- 9) Verify that positive pressure is maintained on nitrogen-blanketed transformers;
 - 10) Integrity and contamination of bus insulation system;
 - 11) Verify that tap-changer is set at correct ratio for rated voltage under normal operating conditions;
 - 12) Verify proper secondary voltage phase-to-phase and phase-to-earth after energisation and prior to loading. This shall be confirmed by Contractor's LEW in liaison with equipment manufacturer;
 - 13) Perform specific inspections and mechanical tests as recommended by manufacturer;
 - 14) Verify correct equipment earthing;
 - 15) Ensure that the name plate information for transformers are mounted on the transformer properly; and
 - 16) Ensure that the laminated Trip and Alarm setting labels for mechanical gauges, instruments or any other measuring devices are provided.
- (ii) Electrical Tests (Shall be carried out by the Contractor's LEW and witnessed by S.O.):
- 1) Insulation Resistance Tests:
 - Applied megohmmeter DC voltage in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards:
 - Winding-to-winding; and
 - Winding-to-earth.
 - 10-minute test duration with resistances tabulated at 30 seconds, 1 minute, and 10 minutes;
 - Results temperature corrected in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards;
 - Temperature corrected insulation resistance values equal to, or greater than, ohmic values established by manufacturer; and
 - Insulation resistance test results to compare within 1 % of adjacent windings.
 - 2) Perform tests and adjustments for fans, controls, and alarm functions as suggested by manufacturer.
 - 3) Perform AC High Pressure tests on transformer windings.
 - 4) Perform turns-ratio tests at all tap positions.
 - 5) Test load tap-changer where applicable in accordance with manufacturer's recommendations.
 - 6) Perform insulation power-factor/dissipation-factor tests for larger sub-station type transformers on windings in accordance with test equipment manufacturer's published data.

- 7) Perform excitation-current tests in accordance with test equipment manufacturer's published data.
 - 8) Perform tests on all instrument transformers.
 - 9) Sample insulating oil in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards and have laboratory tested for:
 - Dielectric breakdown voltage in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards;
 - Acid neutralisation number in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards;
 - Interfacial tension in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards;
 - Colour in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards;
 - Visual condition in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards;
 - Specific gravity in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards;
 - Water content, in parts per million, in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards;
 - Dielectric fluid test results in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards;
 - Power factor at 25 degrees C and at 100 degrees, in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards; and
 - Maximum power factor, corrected to 20 degrees C, in accordance with manufacturer's specifications.
- (d) Low Voltage Cables, 600 V Maximum
- (i) Visual and Mechanical Inspection (Shall be carried out by Contractor's LEW and witnessed by S.O.):
 - 1) Inspect Each Individual Exposed Power Cable for:
 - Physical damage;
 - Proper connections in accordance with single-line diagram;
 - Cable bends not in conformance with manufacturer's minimum allowable bending radius where applicable;
 - Colour coding conformance with specifications;
 - Proper circuit identification; and
 - Proper cable tagging for all incoming and outgoing cables.

- 2) Mechanical Connections for:
 - Proper lug type for conductor material;
 - Proper lug installation; and
 - Bolt torque level in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards unless otherwise specified by manufacturer.
 - 3) Shielded Instrumentation Cables for:
 - Proper shield earthing;
 - Proper terminations;
 - Proper circuit identification; and
 - Proper cable tagging for all incoming and outgoing cables.
 - 4) Control Cables for:
 - Proper termination;
 - Proper circuit identification; and
 - Proper cable tagging for all incoming and outgoing cables.
 - 5) Cables Terminated Through Window Type CTs: Verify that neutrals and earths are terminated for correct operation of protective devices.
- (ii) Electrical Tests for Conductors (Shall be carried out by Contactor's LEW and witnessed by S.O.):
- 1) Insulation Resistance Tests:
 - Utilise 1,000 V DC megohmmeter for 600 V insulated conductors and 500 V DC megohmmeter for 300 V insulated conductors;
 - Test each conductor with respect to earth and to adjacent conductors for 1 minute;
 - Evaluate ohmic values by comparison with conductors of same length and type; and
 - Investigate if the values are less than 50 megohms.
 - 2) Continuity test by ohmmeter method to ensure proper cable connections.
- (e) High Voltage Cables
- (i) Visual and Mechanical Inspection (Shall be carried out by Contractor's LEW and witnessed by S.O.):
 - 1) Inspect Each Individual Exposed Cable for:
 - Physical damage plus jacket and insulation condition;
 - Proper connections in accordance with single-line diagram;
 - Proper shield earthing;
 - Proper cable support;
 - Proper cable termination;

- Cable bends not in conformance with manufacturer's minimum allowable bending radius;
 - Proper arc and fireproofing in common cable areas;
 - Proper circuit and phase identification; and
 - Proper cable tagging for all incoming and outgoing cables.
- 2) Mechanical Connections for:
- Proper lug type for conductor material;
 - Proper lug installation; and
 - Bolt torque level in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI, unless otherwise specified by manufacturers.
- 3) Conductors Terminated Through Window Type CTs: Verify that neutrals and grounds are terminated for correct operation of protective devices.
- (ii) Electrical Tests (Shall be carried out by Contractor's LEW and witnessed by S.O.):
- 1) Insulation Resistance Tests:
- Utilise 2,500 V megohmmeter for 6 kV conductors and 5,000 V megohmmeter for 12 kV and 24 kV conductors;
 - Test each cable individually with remaining cables and shields earthed;
 - Test each conductor with respect to earth and to adjacent conductors in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards;
 - Evaluate ohmic values by comparison with conductors of same length and type; and
 - Investigate values if the values are less than 50 megohms.
- 2) Shield Continuity Tests:
- By ohmmeter method on each section of conductor; and
 - Investigate values in excess of 10 ohms per 300 metres of conductors.
- 3) High Potential Tests:
- In accordance with requirements of applicable Singapore Standards, IEC 60502- Part 2 for high voltage XLPE insulated conductors;
 - Each Conductor Section Tested With:
 - Splices and terminations in-place but disconnected from equipment; and
 - Remaining conductors and shields earthed in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards.
 - Apply maximum test voltage recommended by IEC Standard for voltage class of cable being tested;

- Utilise guard ring or field reduction sphere to suppress corona at disconnected terminations;
 - Maximum test voltage shall not exceed limits for terminators specified in manufacturer's specifications;
 - Test duration shall be 15 minutes or as per IEC-60502 or as specified by the manufacturer;
 - Immediately following test, earth conductor for adequate time period to drain insulation stored charge; and
 - Test results evaluated on a pass/fail basis.
- 4) New Conductors Spliced to Existing Conductors:
- Prior to performing splices, high potential test new conductor sections; and
 - After splicing new conductors to existing conductors, disconnect the existing conductors and perform the following tests:
 - Shield continuity test;
 - Insulation resistance test; and
 - High potential test with test voltage shall be in accordance with IEC-60502 Part 2.
- (f) Disconnectors and Isolators
- (i) Visual and Mechanical Inspection (Shall be carried out by Contractor's LEW and witnessed by S.O.):
- 1) Proper blade pressure and alignment;
 - 2) Proper operation of switch operating handle;
 - 3) Adequate mechanical support for each fuse;
 - 4) Proper contact-to-contact tightness between fuse clip and fuse;
 - 5) Cable connection bolt torque level in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards;
 - 6) Proper phase barrier material and installation;
 - 7) Verify that fuse sizes and types correspond to one-line diagram;
 - 8) Perform mechanical operational test and verify interlocking system operation and sequencing;
- (ii) Electrical Tests (Shall be carried out by Contractor's LEW and witnessed by S.O.):
- 1) Insulation Resistance Tests:
 - Applied megohmmeter DC voltage in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards;
 - Phase-to-phase and phase-to-earth for 1 minute on each pole; and
 - Insulation resistance values equal to, or greater than, ohmic values established by manufacturer.

- 2) Contact Resistance Tests:
 - Contact resistance in microhms across each switch blade and fuse holder; and
 - Investigate deviation of 50 % or more from adjacent poles or similar switches.
- (g) Moulded and Insulated Case Circuit Breakers
 - (i) General: Inspection and testing limited to circuit breakers rated 100 amperes and larger in consumer units, distribution boards and all circuit breakers in switchgear, switchboards, and motor control centres.
 - (ii) Visual and Mechanical Inspection (Shall be carried out by Contractor's LEW and witnessed by S.O.):
 - 1) Proper mounting;
 - 2) Proper conductor size;
 - 3) Feeder designation according to nameplate and one-line diagram;
 - 4) Cracked casings;
 - 5) Connection bolt torque level in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards;
 - 6) Operate breaker to verify smooth operation;
 - 7) Compare frame size and trip setting with circuit breaker schedules or one-line diagram; and
 - 8) Verify that terminals are suitable for 75 degrees C rated insulated conductors.
 - (iii) Electrical Tests (Shall be carried out by Contractor's LEW and witnessed by S.O.):
 - 1) Insulation Resistance Tests:
 - Utilise 1,000 V DC megohmmeter for 415 V circuit breakers and 500 V DC megohmmeter for 250 V circuit breakers;
 - Phase-to-phase and phase-to-earth with breaker contacts opened for 1 minute;
 - Phase-to-phase and phase-to-earth with breaker contacts closed for 1 minute; and
 - Test values to comply with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards.
 - 2) Contact Resistance Tests:
 - Contact resistance in microhms across each pole; and
 - Investigate deviation of 50 % or more from adjacent phases and similar breakers.
 - 3) Primary Current Injection Test to Verify:
 - Long-time minimum pickup and delay;

- Short-time pickup and delay;
 - Earth fault pickup and delay;
 - Instantaneous pickup by run-up or pulse method;
 - Trip characteristics of adjustable trip breakers shall be within manufacturer's published time-current characteristic tolerance band, including adjustment factors;
 - Trip times shall be within limits established by requirements of applicable Singapore Standards, IEC Standards, or BSI Standards; and
 - Instantaneous pickup value shall be within values established by requirements of applicable Singapore Standards, IEC Standards, or BSI Standards.
- (h) Low Voltage Power Circuit Breakers
- (i) Visual and Mechanical Inspection (Shall be carried out by Contractor's LEW and witnessed by S.O.):
- 1) Proper mounting, cell fit, and element alignment;
 - 2) Proper operation of racking interlocks;
 - 3) Check for damaged arc chutes;
 - 4) Proper contact condition;
 - 5) Bolt torque level in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards;
 - 6) Perform mechanical operational and contact alignment tests in accordance with manufacturer's instructions;
 - 7) Check operation of closing and tripping functions of trip devices by activating earth fault relays, under voltage shunt relays, and other auxiliary protective devices;
 - 8) Verify primary and secondary contact wipe, gap setting, and other dimensions vital to breaker operation are correct;
 - 9) Check charging motor, motor brushes, associated mechanism, and limit switches for proper operation and condition;
 - 10) Check operation of electrically operated breakers in accordance with manufacturer's instructions; and
 - 11) Check for adequate lubrication on contact, moving, and sliding surfaces.
- (ii) Electrical Tests (Shall be carried out by Contractor's LEW and witnessed by S.O.):
- 1) Insulation Resistance Tests:
 - Utilise 1,000 V DC megohmmeter for 415 V circuit breakers;
 - Phase-to-phase and phase-to-earth with breaker contacts opened for 1 minute;
 - Phase-to-phase and phase-to-earth with breaker contacts closed for 1 minute; and

- Test values to comply with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards.
- 2) Contact Resistance Tests:
 - Contact resistance in microhms across each pole; and
 - Investigate deviation of 50 % or more from adjacent phases and similar breakers.
- 3) Primary Current Injection Test to Verify:
 - Long-time minimum pickup and delay;
 - Short-time pickup and delay;
 - Earth fault pickup and delay;
 - Instantaneous pickup by run-up or pulse method; and
 - Trip characteristic when adjusted to setting sheet parameters shall be within manufacturer's published time-current tolerance band.
- (i) High Voltage Vacuum Circuit Breakers
 - (i) Visual and Mechanical Inspection (Shall be carried out by Contractor's LEW and witnessed by S.O.):
 - 1) Check for proper element alignment;
 - 2) Check for proper operation of cubicle shutters and racking mechanism;
 - 3) Bolt torque level in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards;
 - 4) Perform mechanical operational tests on breaker and its operating mechanism in accordance with manufacturer's instructions, plus check:
 - Pull rod adjustment;
 - Trip latch clearance;
 - Over travel stops; and
 - Wipe and gap setting.
 - 5) Perform breaker travel and velocity analysis in accordance with manufacturer's instructions; values shall be in accordance with manufacturer's acceptable limits;
 - 6) Check contact erosion indicators in accordance with manufacturer's instructions; and
 - 7) With Breaker in TEST Position:
 - Trip and close breaker with control switch; and
 - Trip breaker by manually operating each protective relay.
 - (ii) Electrical Tests (Shall be carried out by Contractor's LEW and witnessed by S.O.):
 - 1) Insulation Resistance Tests:
 - Utilise 2,500 V DC megohmmeter for all circuit breakers;

- Phase-to-phase and phase-to-earth with breaker contacts opened for 1 minute;
 - Phase-to-phase and phase-to-earth with breaker contacts closed for 1 minute; and
 - Test values to comply with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards.
- 2) Contact Resistance Tests:
- Between the line and load stab of closed contact resistance in microhms across each pole; and
 - Investigate deviation of 50 % or more from adjacent poles and similar breakers.
- 3) Over Potential Tests:
- Maximum applied AC or DC voltage in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards;
 - Each phase-to-earth with other poles grounded and contacts closed for 1 minute; and
 - Test results evaluated on pass/fail basis.
- 4) Minimum pickup voltage tests on trip and close coils.
- 5) Control Wiring Tests:
- Insulation resistance test at 1,000 V DC on control wiring except that connected to solid state components;
 - Insulation resistance to be 1 megaohm minimum;
 - All unused cables are to be labelled as 'Not Used' and bundle together with proper cable ties. No exposed unused cables are not allowed. Contractor shall submit the termination drawings to S.O. for verification;
 - Contractor shall ensure that all wirings and components are properly labelled and terminated; and
 - The Contractor shall provide detailed termination drawings to S.O. for checking and verification.
- 6) Vacuum bottle over potential integrity test across each vacuum bottle with breaker in OPEN position, in accordance with manufacturer's instructions.
- 7) Power Factor Test on Each Phase:
- With breaker in both OPEN and CLOSED position; and
 - Compare power factor and arc chute watt loss with adjacent poles or manufacturer's published data.

- 8) Power Factor Test on Each Bushing:
 - Utilise conductive straps and hot collar procedures if bushings are not equipped with power factor tap; and
 - Power factor and capacitance test results within nameplate rating of bushings.
- (j) High Voltage SF₆ Circuit Breakers (66 kV)
 - (i) Visual and Mechanical Inspection (Shall be carried out by Contractor's LEW and witnessed by S.O.):
 - 1) Compare equipment nameplate data with Drawings and Specifications.
 - 2) Inspect physical and mechanical condition.
 - 3) Confirm correct application of manufacturer's recommended lubricants.
 - 4) Inspect anchorage and grounding.
 - 5) Inspect and verify adjustments of mechanism in accordance with manufacturer's published data.
 - 6) Inspect air compressor in accordance with manufacturer's published data.
 - 7) Inspect hydraulic system in accordance with manufacturer's published data.
 - 8) Test for gas leaks.
 - 9) Verify correct operation of all air and SF₆ gas pressure alarms and cut-outs.
 - 10) Slow close/open breaker and check for binding.
 - 11) Perform time-travel analysis.
 - 12) Test SF₆ gas for moisture and SO₂ content.
 - 13) Inspect all bolted electrical connections for high resistance using one of the following methods:
 - Use of low-resistance ohmmeter in accordance with relevant standards;
 - Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data; and
 - Perform thermographic survey.
 - 14) Record as-found and as-left counter operations.
 - 15) Check spring charging motor starts counter and compare with operations counter, if applicable.
 - 16) Verify operation of all heaters.
 - (ii) Electrical Tests (Shall be carried out by Contractor's LEW and witnessed by S.O.):
 - 1) Perform a contact resistance test.
 - 2) Perform resistance measurements through all bolted connections with low-resistance ohmmeter, if applicable.

- 3) Perform insulation-resistance tests pole-to-pole, pole-to-ground, and across open poles at 15,000 V minimum.
 - 4) Perform insulation-resistance tests on all control wiring at 1,000 V DC. For units with solid-state components, follow manufacturer's recommendations.
 - 5) Perform dissipation-factor/power-factor tests on breaker and bushings.
 - 6) Perform over potential test in accordance with manufacturer's instructions.
 - 7) Perform minimum pick-up voltage test on trip and close coils.
 - 8) Verify trip, close, trip-free, and anti-pump functions.
 - 9) Trip circuit breaker by operation of each protective device.
- (iii) Test Values:
- 1) Compare bolted connection resistance to values of similar connections.
 - 2) Bolt-torque levels should be in accordance with relevant standards unless otherwise specified by manufacturer.
 - 3) Microhm or millivolt drop values shall not exceed the high levels of the normal range as indicated in the manufacturer's published data. If manufacturer's data is not available, investigate any values which deviate from similar connections by more than 50 % of the lowest value.
 - 4) Compare time-travel data with manufacturer's published data.
 - 5) Control wiring insulation resistance should be a minimum of 2 megohms.
 - 6) Circuit breaker insulation resistance should be in accordance with relevant standards.
 - 7) Dissipation-factor/power-factor test results shall be compared to manufacturer's published data. In the absence of manufacturer's published data, the comparison shall be made to similar breakers.
 - 8) Dissipation-factor/power-factor and capacitance test results should be within ten % of nameplate rating for bushings.
 - 9) The insulation shall withstand the over potential test voltage applied
 - 10) Minimum pickup for trip and close coils shall conform to manufacturer's published data.
- (k) Protective Relays
- (i) Visual and Mechanical Inspection (Shall be carried out by Contractor's LEW and witnessed by S.O.):
 - 1) Visually Check Each Relay for:
 - Tight cover gasket and proper seal;
 - Unbroken cover glass;
 - Condition of spiral spring and contacts;
 - Disc clearance; and

- Condition of case shorting contacts if present.
- 2) Mechanically Check Each Relay for:
 - Freedom of movement; and
 - Proper travel and alignment.
- 3) Verify that Each Relay:
 - Complies with Specifications and application; and
 - Is set in accordance with recommended settings.
- (ii) Electrical Tests (Shall be carried out by Contractor's LEW and witnessed by S.O.):
 - 1) Insulation resistance test on each circuit to frame except for solid state devices.
 - 2) Tests on Nominal Recommended Setting for:
 - Pickup parameters on each operating element;
 - Timing at three points on time-current curve;
 - Pickup target and seal-in units; and
 - Special tests as required to check operation of restraint, directional, and other elements in accordance with manufacturer's instruction manual.
 - 3) Phase angle and magnitude contribution tests on differential and directional relays after energisation to vectorially verify proper polarity and connections.
 - 4) Test all microprocessor type protective relays in full accordance with manufacturer's instructions.
 - 5) Current Injection Tests:
 - For entire current circuit in each section;
 - Secondary injection currents as recommended by manufacturer;
 - Test current at each device; and
 - Relays shall be tested for accuracy and compliance with relevant standards by the secondary injection method.
 - 6) DC element test is to be performed on auxiliary relays to check pick-up voltage and contact resistance.
- (I) Instrument Transformers
 - (i) Visual and Mechanical Inspection (Shall be carried out by Contractor's LEW and witnessed by S.O.):
 - 1) Visually Check Current, Potential, and Control Transformers for:
 - Cracked insulation;
 - Broken leads or defective wiring;
 - Proper connections; and

- Adequate clearances between primary and secondary circuit wiring.
- 2) Verify Mechanically that:
 - Earthing and shorting connections have good contact; and
 - Withdrawal mechanism and grounding operation, when applicable, operate properly.
- 3) Verify proper primary and secondary fuse sizes for potential transformers.
- (ii) Electrical Tests (Shall be carried out by Contractor's LEW and witnessed by S.O.):
 - 1) Current Transformer Tests:
 - Insulation resistance test of transformer and wiring-to-ground at 1,000 V DC for 30 seconds;
 - Polarity test; and
 - Primary injection test:
 - Perform an excitation test on transformers used for relaying applications in accordance with IEC; and
 - The magnetization (excitation) curve of all transformers for protection schemes shall be verified at site.
 - 2) Potential Transformer Tests:
 - Insulation resistance test at test voltages in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards:
 - Winding-to-winding; and
 - Winding-to-earth.
 - Polarity test to verify polarity marks or H1-X1 relationship as applicable; and
 - The magnetization (excitation) curve of all transformers for protection schemes shall be verified at site.
 - 3) Insulation resistance measurement on instrument transformer shall not be less than that shown in requirements of applicable Singapore Standards, IEC Standards, or BSI Standards.
- (m) Metering
 - (i) Visual and Mechanical Inspection (Shall be carried out by Contractor's LEW and witnessed by S.O.):
 - 1) Verify metre connections in accordance with appropriate diagrams;
 - 2) Verify metre multipliers;
 - 3) Verify that metre types and scales conform to Specifications and Drawings;
 - 4) Check calibration of metres at cardinal points; and
 - 5) Check calibration of electrical transducers where shown on Drawings.

- (n) Earthing Systems (System Earth)
- (i) Visual and Mechanical Inspection (Shall be carried out by Contractor's LEW and witnessed by S.O.):
- 1) Equipment and circuit earth connections in motor control centres, distribution boards, switchboard, and switchgear assemblies for proper connection and tightness.
 - 2) Earth bus connections in motor control centres, distribution boards, switchboard, and switchgear assemblies for proper termination and tightness.
 - 3) Effective transformer core and equipment earthing.
 - 4) Accessible connections to earthing electrodes for proper fit and tightness.
 - 5) Accessible exothermic-weld earthing connections to verify that moulds were fully filled and proper bonding was obtained.
- (ii) Electrical Tests (Shall be carried out by Contractor's LEW and witnessed by S.O.):
- 1) Fall-of-Potential Test:
 - In accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards for measurement of main earth system's resistance; and
 - Main earth electrode system resistance to earth to be no greater than 1 ohm.
 - 2) Two-Point Direct Method Test:
 - In accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards for measurement of earth resistance between main earth system, equipment frames, and system neutral and derived neutral points; and
 - Equipment earth resistance shall not exceed main earth system resistance by 0.25 ohm.
- (o) Earthing Systems (Instrument Earth)
- (i) Visual and Mechanical Inspection (Shall be carried out by Contractor's LEW and witnessed by S.O.):
- 1) Connections in earth bar for tightness;
 - 2) Earth bus connections in instrument panels for proper termination and tightness;
 - 3) Accessible connections to earthing electrodes for proper fit and tightness; and
 - 4) Accessible exothermic-weld earthing connections to verify that moulds were fully filled and proper bonding was obtained.

- (ii) Electrical Tests (Shall be carried out by Contractor's LEW and witnessed by S.O.):
 - 1) Fall-of-Potential Test:
 - In accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards for measurement of main earth system's resistance; and
 - Main earth electrode system resistance to earth to be no greater than 1 ohm.
- (p) Earthing Systems (Lightning Protection)
 - (i) Visual and Mechanical Inspection (Shall be carried out by Contractor's LEW and witnessed by S.O.):
 - 1) Accessible connections to earthing electrodes for proper fit and tightness; and
 - 2) Accessible exothermic-weld earthing connections to verify that moulds were fully filled and proper bonding was obtained.
 - (ii) Electrical Tests (Shall be carried out by Contractor's LEW and witnessed by S.O.):
 - 1) Fall-of-Potential Test:
 - In accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards for measurement of main earth system's resistance; and
 - Main earth electrode system resistance to earth to be no greater than 10 ohms.
- (q) Earth Fault Systems
 - (i) Inspection and Testing Limited to:
 - 1) Zero sequence earthing systems; and
 - 2) Residual earth fault systems.
 - (ii) Visual and Manual Inspection (Shall be carried out by Contractor's LEW and witnessed by S.O.):
 - 1) Neutral Main Bonding Connection to Assure:
 - Zero sequence sensing system is earthed ahead of neutral disconnect link;
 - Earth strap sensing system is earthed through sensing device; and
 - Neutral earth conductor is solidly earthed.
 - 2) Verify that control power has adequate capacity for system.
 - 3) Manually Operate Monitor Panels for:
 - Trip test;
 - No trip test; and
 - Non-automatic reset.

- 4) Zero sequence system for symmetrical alignment of core balance transformers about current carrying conductors.
 - 5) Relay check for pickup and time under simulated earth fault conditions.
 - 6) Verify nameplate identification by device operation.
- (iii) Electrical Tests (Shall be carried out by Contractor's LEW and witnessed by S.O.):
- 1) Test system neutral insulation resistance with neutral earth link removed. System neutral insulation resistance minimum 1 megohm.
 - 2) Determine relay pickup by primary current injection at the sensor. Relay pickup current within plus or minus 10 % of device dial or fixed setting.
 - 3) Test relay timing by secondary injection test using values recommended by manufacturer. Relay operating time in accordance with manufacturer's time-current characteristic curves.
 - 4) Test system operation at 55 % rated control voltage, if applicable.
 - 5) Test zone interlock system by simultaneous sensor current injection and monitoring zone blocking functions.
- (r) AC Induction Motors
- (i) Visual and Mechanical Inspection (Shall be carried out by Contractor's LEW and witnessed by S.O.):
- 1) Proper electrical and earth connections.
 - 2) Shaft alignment.
 - 3) Blockage of ventilating air passageways.
 - 4) Operate Motor and Check for:
 - Excessive mechanical and electrical noise;
 - Overheating;
 - Correct rotation;
 - Check vibration detectors, resistance temperature detectors, or motor inherent protectors for functionality and proper operation; and
 - Excessive vibration.
 - 5) Check operation of space (condensation) heaters.
- (ii) Electrical Tests (Shall be carried out by Contractor's LEW and witnessed by S.O.):
- 1) Insulation Resistance Tests:
 - At test voltages established by requirements of applicable Singapore Standards, IEC Standards, or BSI Standards for:
 - Motors rated 150 kW and larger for 10-minute duration with resistances tabulated at 30 seconds, 1 minute, and 10 minutes; and
 - Motors rated less than 150 kW for 1-minute duration with resistances tabulated at 30 and 60 seconds.

- Insulation resistance values equal to, or greater than, ohmic values established by manufacturers.
 - 2) Calculate polarisation index ratios for motors above 149 kilowatts. Investigate index ratios less than 1.5 for Class A insulation and 2.0 for Class B insulation.
 - 3) Insulation resistance test on insulated bearings in accordance with manufacturer's instructions.
 - 4) Measure running current and voltage, and evaluate relative to load conditions and nameplate full-load amperes.
 - 5) For Slip-ring induction motors, additionally, test for correct operation of brush lifting device, slip-ring shorting device, correct start up sequence of rotor resistance controller and its operation time and thermal protection, reset to starting position at end of starting.
 - 6) Over potential Tests:
 - Applied DC voltage in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards;
 - Limited to high voltage motors; and
 - Test results evaluated on pass/fail basis.
- (s) High Voltage Motor Control Equipment
- (i) Visual and Mechanical Inspection:
- 1) Insulator damage and contaminated surfaces;
 - 2) Proper barrier and shutter installation and operation;
 - 3) Proper operation of indicating devices;
 - 4) Proper overload protection;
 - 5) Blockage of air cooling passages;
 - 6) Proper operation of draw out elements;
 - 7) Integrity and contamination of bus insulation system;
 - 8) Check Door and Device Interlocking System by:
 - Closure attempt of device when door is in OFF or OPEN position; and
 - Opening attempt of door when device is in ON or CLOSED position.
 - 9) Check Nameplates for Proper Identification of Each:
 - Equipment title and tag number with latest single-line diagram;
 - Pushbutton;
 - Control switch;
 - Pilot light;
 - Control relay;
 - Circuit breaker;
 - Indicating metre;

- E-Stop Push Buttons;
 - Protection relays; and
 - Verify that fuse sizes and types conform to Contract Documents.
- 10) Check Connections for High Resistance by Low Resistance Ohmmeter and Thermographic Survey:
- Ohmic value to be zero; and
 - Thermographic survey temperature gradient of 2 degrees C, or less.
- 11) Verify performance of each control device and feature supplied as part of the motor controllers.
- 12) Control Wiring:
- Compare wiring to local and remote control, and protective devices with elementary diagrams;
 - Check for proper conductor lacing and bundling;
 - Check for proper conductor identification;
 - Check for proper conductor lugs and connections;
 - Contractor shall ensure that all wirings and components are properly labelled and terminated;
 - The Contractor shall provide detailed termination drawings to S.O. for checking and verification;
 - All unused cables are to be labelled as 'Not Used' and bundle together with proper cable ties; and
 - Exposed unused cables are not allowed.
- 13) Exercise active components.
- 14) Inspect Magnetic Contactors for:
- Correct mechanical operations;
 - Correct contact gap, wipe, alignment, and pressure; and
 - Correct torque of connections.
- (ii) Electrical Tests:
- 1) Insulation Resistance Tests:
- Applied megohmmeter DC voltage in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards;
 - Contactor phase-to-earth and across open contacts for 1 minute on each phase;
 - Starter section phase-to-phase and phase-to-earth on each phase with starter contacts closed and protective devices open; and
 - Test values to comply with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards.

- 2) Bottle integrity test for vacuum contactors in accordance with manufacturer's procedure.
 - 3) Test by Primary Current Injection:
 - Overload units at sensors using 300 % of motor full-load current: Overload trip times to be in accordance with manufacturer's published data; and
 - Check voltage levels at each point on terminal boards and each device terminal.
 - 4) Control Wiring Tests:
 - Apply secondary voltage on control power and potential circuits;
 - Check voltage levels at each point on terminal boards and at each device terminal;
 - Contractor shall ensure that all wirings and components are properly labelled and terminated;
 - The Contractor shall provide detailed termination drawings to S.O. for checking and verification; and
 - Insulation resistance test at 1,000 V DC on control wiring except that connected to solid state components: Insulation resistance to be 1 megohm minimum.
 - 5) Test indicating and monitoring devices for proper operation.
 - 6) Perform set-up and testing of solid state relays and multifunction protective devices in accordance with manufacturer's instructions.
 - 7) Measure Contact and Power Fuse Resistances:
 - Contact resistance shall not exceed manufacturer's recommended values;
 - Resistance of power fuses not to deviate more than 15 % between identical fuses;
 - All unused cables are to be labelled as 'Not Used' and bundle together with proper cable ties; and
 - Exposed unused cables are not allowed.
- (t) Low Voltage Motor Control
- (i) Visual and Mechanical Inspection:
 - 1) Proper barrier and shutter installation and operation;
 - 2) Proper operation of indicating and monitoring devices;
 - 3) Proper overload protection for each motor;
 - 4) Improper blockage of air cooling passages;
 - 5) Proper operation of draw out elements;
 - 6) Integrity and contamination of bus insulation system;

- 7) Check Door and Device Interlocking System by:
 - Closure attempt of device when door is in OFF or OPEN position; and
 - Opening attempt of door when device is in ON or CLOSED position.
- (ii) Check Key Interlocking Systems for:
 - 1) Key captivity when device is in ON or CLOSED position;
 - 2) Key removal when device is in OFF or OPEN position;
 - 3) Closure attempt of device when key has been removed;
 - 4) Correct number of keys in relationship to number of lock cylinders; and
 - 5) Existence of other keys capable of operating lock cylinders; destroy duplicate sets of keys.
- (iii) Check Nameplates for Proper Identification of:
 - 1) Equipment title and tag number with latest one-line diagram;
 - 2) Pushbuttons;
 - 3) Control switches;
 - 4) Pilot lights;
 - 5) Control relays;
 - 6) Circuit breakers;
 - 7) Indicating metres; and
 - 8) Verify that fuse and circuit breaker sizes and types conform to Drawings.
- (iv) Verify that current and voltage transformer ratios conform to Drawings.
- (v) Check Bus Connections for High Resistance by Low Resistance Ohmmeter and Thermographic Survey:
 - 1) Ohmic value to be zero; and
 - 2) Thermographic survey temperature gradient of 2 degrees C, or less.
- (vi) Check Operation and Sequencing of Electrical and Mechanical Interlock Systems by:
 - 1) Closure attempt for locked open devices;
 - 2) Opening attempt for locked closed devices; and
 - 3) Key exchange to operate devices in OFF-NORMAL positions.
- (vii) Verify performance of each control device and feature supplied as part of the motor control centre.
- (viii) Control Wiring:
 - 1) Compare wiring to local and remote control, and protective devices with elementary diagrams;
 - 2) Check for proper conductor lacing and bundling;
 - 3) Check for proper conductor identification;

- 4) Check for proper conductor lugs and connections:
 - All unused cables are to be labelled as 'Not Used' and bundle together with proper cable ties;
 - Exposed unused cables are not allowed;
 - Contractor shall ensure that all wirings and components are properly labelled and terminated; and
 - The Contractor shall provide detailed termination drawings to S.O. for checking and verification.
- (ix) Exercise active components.
- (x) Inspect Contactors for:
 - 1) Correct mechanical operations;
 - 2) Correct contact gap, wipe, alignment, and pressure; and
 - 3) Correct torque of all connections.
- (xi) Compare overload heater rating with full-load current for proper size.
- (xii) Compare motor protector and circuit breaker with motor characteristics and power factor correction capacitors for proper size.
- (xiii) Perform phasing check on double-ended motor control centres to ensure proper bus phasing from each source.
- (u) Electrical Tests (Shall be carried out by Contractor's LEW and witnessed by S.O.):
 - (i) Insulation Resistance Tests:
 - 1) Applied megohmmeter DC voltage in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards;
 - 2) Bus section phase-to-phase and phase-to-earth for 1 minute on each phase;
 - 3) Contactor phase-to-earth and across open contacts for 1 minute on each phase;
 - 4) Starter section phase-to-phase and phase-to-earth on each phase with starter contacts closed and protective devices open; and
 - 5) Test values to comply with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards.
 - (ii) Over potential Tests:
 - 1) Maximum applied AC or DC voltage in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards; and
 - 2) Phase-to-phase and phase-to-earth for 1 minute for each phase of each bus section; and
 - 3) Test results evaluated on pass/fail basis.

- (iii) Current Injection through Overload Unit at 300 % of Motor Full-Load Current and Monitor Trip Time:
 - 1) Trip time in accordance with manufacturer's published data, and
 - 2) Investigate values in excess of 120 seconds.
- (iv) Control Wiring Tests:
 - 1) Apply secondary voltage to control power and potential circuits;
 - 2) Check voltage levels at each point on terminal boards and each device terminal; and
 - 3) Insulation resistance test at 1,000 V DC on control wiring except that connected to solid state components: Insulation resistance to be 1 megaohm minimum.
 - All unused cables are to be labelled as 'Not Used' and bundle together with proper cable ties;
 - Exposed unused cables are not allowed;
 - Contractor shall ensure that all wirings and components are properly labelled and terminated; and
 - The Contractor shall provide detailed termination drawings to S.O. for checking and verification.
- (v) Operational test by initiating control devices to affect proper operation.
- (v) Low Voltage Motor Starters
 - (i) Perform insulation resistance test for all the control circuit.
 - (ii) Perform testing on the control circuit for local and remote operation.
 - (iii) Perform process interlock testing (if any).
 - (iv) Perform testing on the sequence of operation of all conductors.
 - (v) Perform testing on the indication and signals circuits.
- (w) Automatic Transfer Switches
 - (i) Visual and Mechanical Inspection:
 - 1) Check doors and panels for proper interlocking.
 - 2) Check connections for high resistance by low resistance ohmmeter and thermographic survey.
 - 3) Check positive mechanical and electrical interlock between normal and alternate sources.
 - 4) Check for Proper Operation:
 - Manual transfer function switch;
 - Generator under load and non-load conditions;
 - Auto-exerciser of generator under load and no-load conditions; and
 - Transfer switch bypass-isolation switch functions.

- 5) Verify settings and operation of control devices.
- (ii) Electrical Tests (Shall be carried out by Contractor's LEW and witnessed by S.O.):
 - 1) Insulation Resistance Tests:
 - Applied megohmmeter DC voltage in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards for each phase with switch CLOSED in both source positions;
 - Phase-to-phase and phase-to-earth for 1 minute; and
 - Test values in accordance with manufacturer's published data.
 - 2) Contact Resistance Test:
 - Contact resistance in microhms across each switch blade for both source positions, including bypass-isolation switch;
 - Investigate values exceeding 500 microhms; and
 - Investigate values deviating from adjacent pole by more than 50 %.
 - 3) Set and Calibrate in Accordance with Specifications:
 - Voltage and frequency sensing relays;
 - Time delay relays; and
 - Engine start and shutdown relays.
 - 4) Perform Automatic Transfer Tests by:
 - Simulating loss of normal power;
 - Return to normal power;
 - Simulating loss of alternate power; and
 - Simulating single-phase conditions for normal and alternate sources.
 - 5) Monitor and Verify Operation and Timing of:
 - Normal and alternate voltage sensing relays;
 - Engine start sequence;
 - Timing delay upon transfer and retransfer;
 - Engine cool down and shutdown;
 - Interlocks and limit switch functions; and
 - Engine cool down and shutdown feature.
- (x) Battery System
 - (i) Visual and Mechanical Inspection (Shall be carried out by Contractor's LEW and witnessed by S.O.):
 - 1) Physical damage and electrolyte leakage;
 - 2) Evidence of corrosion;
 - 3) Intercell bus link integrity;

- 4) Battery cable insulation damage and contaminated surfaces;
 - 5) Operating conditions of ventilating equipment; and
 - 6) Visual check of electrolyte level.
- (ii) Electrical Tests (Shall be carried out by Contractor's LEW and witnessed by S.O.):
- 1) Measure:
 - Bank charging voltage;
 - Individual cell voltage;
 - Electrolyte specific gravity in each cell; and
 - Measured test values to be in accordance with manufacturer's published data.
 - 2) Verify During Recharge Mode:
 - Charging rates from charger; and
 - Individual cell acceptance of charge.
 - 3) Load tests for integrity and capacity; test values in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards.
- (y) Low Voltage Surge Arrestors
- (i) Visual and Mechanical Inspection (Shall be carried out by Contractor's LEW and witnessed by S.O.):
- 1) Adequate clearances between arrestors and enclosures; and
 - 2) Earth connections to earth bus or electrode.
- (ii) Electrical Tests (Shall be carried out by Contractor's LEW and witnessed by S.O.):
- 1) Varistor Type Arrestors:
 - Clamping voltage test;
 - Rated rms voltage test;
 - Rated DC voltage test; and
 - Varistor arrestor test values in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards.
- (z) Medium Voltage Surge Arrestors and Surge Capacitors
- (i) Visual Inspection (Shall be carried out by Contractor's LEW and witnessed by S.O.):
- 1) Earth connections to earth bus or electrode; and
 - 2) Shortest practical jumper connections to line.
- (ii) Electrical Tests (Shall be carried out by Contractor's LEW and witnessed by S.O.):
- 1) Earthing electrode resistance test in accordance with requirements of applicable Singapore Standards, IEC Standards, or BSI Standards, using three-point fall-of-potential method;

- 2) Insulation power factor;
 - 3) Insulation resistance;
 - 4) RF noise test using Standard Noise Test set with applied voltage of 1.18 times maximum continuous operating voltage;
 - 5) Insulation power factor leakage current, watts loss, and insulation resistance tests in accordance with manufacturer's test values. RIV value not to exceed 10 microV above background noise; and
 - 6) Leakage current and watts loss tests.
- (aa) Engine Driven Standby and Emergency Generator Systems
- (i) Visual and Mechanical Inspection (Shall be carried out by Contractor's LEW and witnessed by S.O.):
 - 1) Proper earthing;
 - 2) Blockage of ventilating passageways;
 - 3) Proper operation of jack water heaters;
 - 4) Integrity of engine cooling and fuel supply systems;
 - 5) Excessive mechanical and electrical noise;
 - 6) Overheating of engine or generator;
 - 7) Proper installation of vibration isolators;
 - 8) Proper cooling liquid type and level;
 - 9) Operate Engine-Generator and Check for:
 - Excessive mechanical and electrical noise;
 - Overheating;
 - Correct rotation;
 - Check resistance temperature detectors or generator inherent thermal protectors for functionality and proper operation; and
 - Excessive vibration.
 - 10) Verify that voltage regulator and governor operation will cause unit speed and output voltage to stabilise at proper values within reasonable length of time;
 - 11) Proper operation of metres and instruments;
 - 12) Compare generator nameplate rating and connection with single-line diagram; and
 - 13) Verify engine-generator operation with variable frequency drives and power factor correction capacitors energised and operating under normal load conditions.

- (ii) Electrical and Mechanical Tests (Shall be carried out by Contractor's LEW and witnessed by S.O.):
- 1) Cold start test by interrupting normal power source with test load consisting of connected building load to verify:
 - Transfer switch operation;
 - Automatic starting operation;
 - Operating ability of engine-generator; and
 - Overcurrent devices capability to withstand inrush currents.
 - 2) Phase rotation tests.
 - 3) Test Engine Protective Shutdown Features for:
 - Low oil pressure;
 - Over temperature; and
 - Over speed.
 - 4) Vibration base-line test on generator sets rated above 250 kW; levels in accordance with manufacturer's recommendations.
 - 5) Load bank test with reactors and resistors adjusted to 80 % power factor for each load step. Record voltage, frequency, load current, oil pressure, and engine coolant temperature at 15-minute intervals:
 - 25 % applied load for 30 minutes;
 - 50 % applied load for 30 minutes;
 - 75 % applied load for 30 minutes;
 - 100 % applied load for 3 hours;
 - 110 % applied load for 30 minutes, continue with 100 % applied load for 1 hour; and
 - Load test results to demonstrate ability of unit to deliver rated load for test period.
 - 6) One-Step Rated kW Load Pickup Test:
 - Perform test immediately after performing load bank test;
 - Apply rated load at 80 % power factor, minus largest rated kW motor, to generator;
 - Start largest rated motor and record voltage drop for 20 cycles minimum with high-speed chart recorder or digital storage oscilloscope; and
 - Compare voltage drop with maximum allowable voltage dip for specified starting situation.

- 7) One-Step Load Rejection Test:
 - Perform test immediately after performing load pickup test;
 - Drop off rated load at 80 % power factor in one step; and
 - Compare voltage drop with maximum allowable voltage rise for specified drop off situation.
- (bb) Uninterruptible Power Systems
 - (i) Visual and Mechanical Inspection (Shall be carried out by Contractor's LEW and witnessed by S.O.):
 - 1) Compare equipment nameplate information with Drawings and specifications;
 - 2) Inspect physical and mechanical condition;
 - 3) Check for correct anchorage, required area clearances, and alignment;
 - 4) Verify that circuit breaker and fuse sizes and types correspond to Drawings;
 - 5) Test all electrical and mechanical interlock systems for correct operation and sequencing;
 - 6) Inspect all bolted electrical connections for high resistance using one of the following methods:
 - Use of low-resistance ohmmeter in accordance with paragraph Electrical Tests;
 - Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or relevant standards; and
 - Perform thermographic survey.
 - 7) Check operation of forced ventilation; and
 - 8) Verify that filters are in place and/or vents are clear.
 - (ii) Electrical Tests (Shall be carried out by Contractor's LEW and witnessed by S.O.):
 - 1) Perform resistance measurements through all bolted connections with low-resistance ohmmeter;
 - 2) Test static transfer from inverter to bypass and back. Use normal load, if possible;
 - 3) Set free-running frequency of oscillator;
 - 4) Test DC under voltage trip level on inverter input breaker. Set according to manufacturer's published data;
 - 5) Test alarm circuits;
 - 6) Demonstrate operation of all metering and control functions;
 - 7) Verify sync indicators for static switch and bypass switches;
 - 8) Demonstrate operation of paralleling function;

- 9) Perform electrical tests for UPS system breakers in accordance with Article Moulded and Insulated Case Circuit Breakers;
 - 10) Perform electrical tests for UPS system automatic transfer switches in accordance with Article Automatic Transfer Switches; and
 - 11) Perform electrical tests for UPS system batteries in accordance with Article Battery System.
- (iii) Test Values:
- 1) Compare bolted connection resistance to values of similar connections;
 - 2) Bolt-torque levels should be in accordance with relevant standards, unless otherwise specified by manufacturer; and
 - 3) Microhm or millivolt drop values shall not exceed the high levels of the normal range as indicated in the manufacturer's published data. If manufacturer's data is not available, investigate any values which deviate from similar connections by more than 50 % of the lowest value.
- (cc) Thermographic Survey (Shall be carried out by Contractor's LEW and witnessed by S.O.)
- (i) Provide a thermographic survey of connections associated with all high voltage cables and all low voltage cables 10 mm² and larger at each:
 - 1) High and low voltage switchgear;
 - 2) Switchboard;
 - 3) Transformer;
 - 4) High voltage starter;
 - 5) Low voltage motor control centre;
 - 6) Distribution board and consumer boards;
 - 7) Variable frequency drive unit;
 - 8) Motors rated 11 kW and larger;
 - 9) Transfer switches;
 - 10) Engine-generators;
 - 11) Disconnectors and isolators; and
 - 12) All other connections not specifically listed, including any splices allowed by S.O..
 - (ii) Remove necessary enclosure metal panels and covers prior to performing survey.
 - (iii) Perform with equipment energised during periods of maximum possible loading.
 - (iv) Do not perform survey on equipment operating at less than 20 % of rated connected operating load.

- (v) Utilise Thermographic Equipment Capable of:
 - 1) Detecting emitted radiation;
 - 2) Converting detected radiation to visual signal; and
 - 3) Detecting 1 degree C temperature difference between subject area and reference point of 30 degrees C.
- (vi) Temperature Gradients of:
 - 1) 3 degrees C to 7 degrees C indicates possible deficiency that warrants investigation;
 - 2) 7 degrees C to 15 degrees C indicates deficiency that is to be corrected as time permits; and
 - 3) 16 degrees C and above indicates deficiency that is to be corrected immediately.
- (vii) Provide Written Report of:
 - 1) Areas surveyed and the resultant temperature gradients;
 - 2) Locations of areas having temperature gradients of 3 degrees C or greater;
 - 3) Cause of heat rise and actions taken to correct the cause of heat rise; and
 - 4) Detected phase unbalance.

END OF SECTION